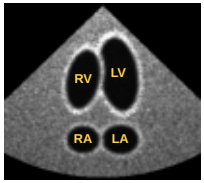
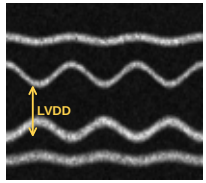
a Input data



2-D echo, apical 4-chamber



M-mode: FS, EPSS, LVDD

Images simulated for illustration; dataset stores derived measurements

T (mo)	δ	age	FS	EPSS	LVDD	WMI
11	1	71	0.26	9	4.6	1
19	1	72	0.38	6	4.1	1.7
16	1	55	0.26	4	3.42	1
57	1	60	0.253	12.1	4.6	1.45
…	…	…	…	…	…	…

UCI Echocardiogram, n = 132 post-MI records (doi:10.24432/C5QW24)

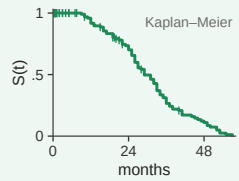
b Reconstruction & audit



Endpoint	all-cause death after MI
T	follow-up time, months
δ = 1	death (n = 88)
δ = 0	right-censored (n = 43)
Checks	schema, ranges, missingness

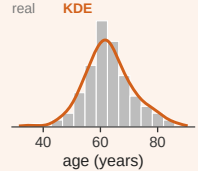
c Modelling arms

Real data · primary



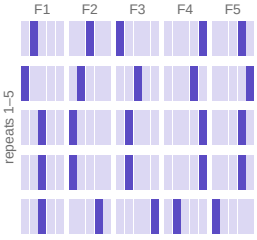
- Cox PH
- FP-Cox (FP2)
- Nested FP-Cox
- Random Survival Forest
- Bootstrap optimism

Synthetic · secondary, training folds only



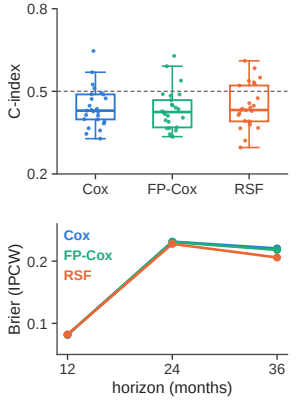
- KDE generation
- Fidelity: SMD, KS
- Bandwidth sweep
- Augment 1×–10×

d Repeated validation

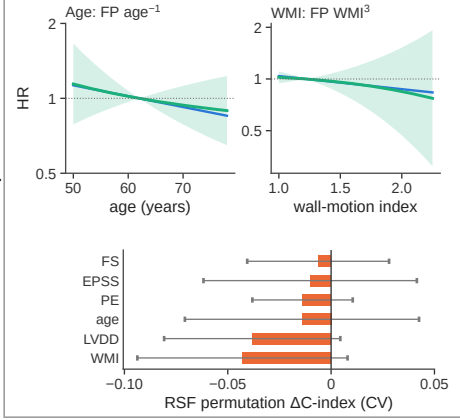


- 5-fold CV × 5 repeats
- 25 held-out test folds
- synthetic rows never tested

e Performance



f Effects & importance



g Output



Data · Real-data analysis · Synthetic experiments · Validation & evaluation · Output